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Design and Implementation of a Buck Converter for DC Voltage Regulation

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1 Introduction and Functionality

DC voltage regulation is vital for delivering stable power to sensitive electronics. While simple, linear regulators waste excessive energy as heat. This project implements a step-down (Buck) Switch-Mode Power Supply (SMPS) to efficiently convert a 15 V input down to a regulated 3 V to 12 V output at a 1 A load current.

1.1 System Operational Overview

As illustrated in Figure 1, the system maintains a stable output voltage via a continuous closed-loop feedback mechanism consisting of four main stages:

- **Power Stage:** Uses an IRF3205 MOSFET switch and a 1N5819 freewheeling diode alternating at 18 kHz to chop the input voltage. A second-order LC low-pass filter smooths this chopped signal into a clean DC output.
- **Feedback Network (K):** A resistive voltage divider continuously samples the output voltage (V_{OUT}) and scales it down to safe signal levels for the control circuitry.
- **Error Amplifier ($A(s)$):** An NE5532 op-amp compares the scaled feedback voltage against a highly stable 2.495 V reference from an LM431 regulator. It amplifies the error and provides compensation to stabilize the loop and eliminate steady-state error.
- **PWM Modulator (F_m):** Compares the amplified error with a locally generated 20 kHz triangular wave to create a variable duty cycle (D) pulse-width modulated signal. This signal is sent to the IR2110 gate driver to turn the power MOSFET on and off.

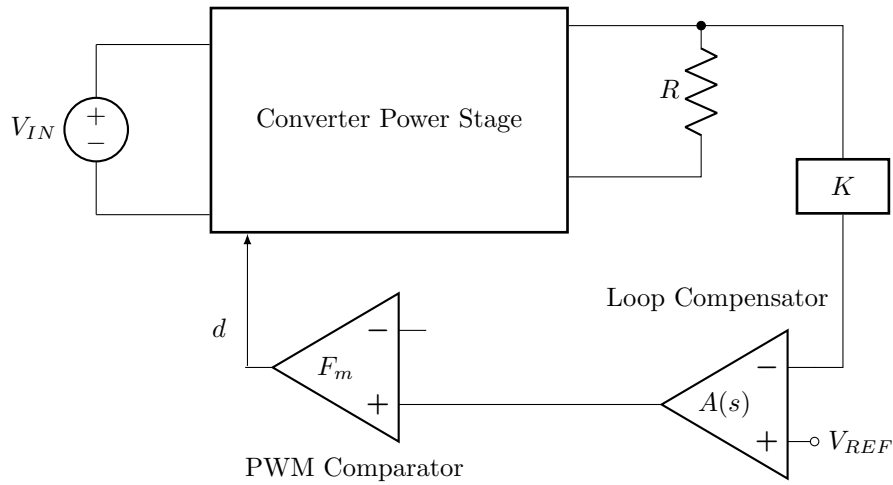


Figure 1: Block Diagram of the System

2 System Architecture

2.1 Triangular Waveform Generation

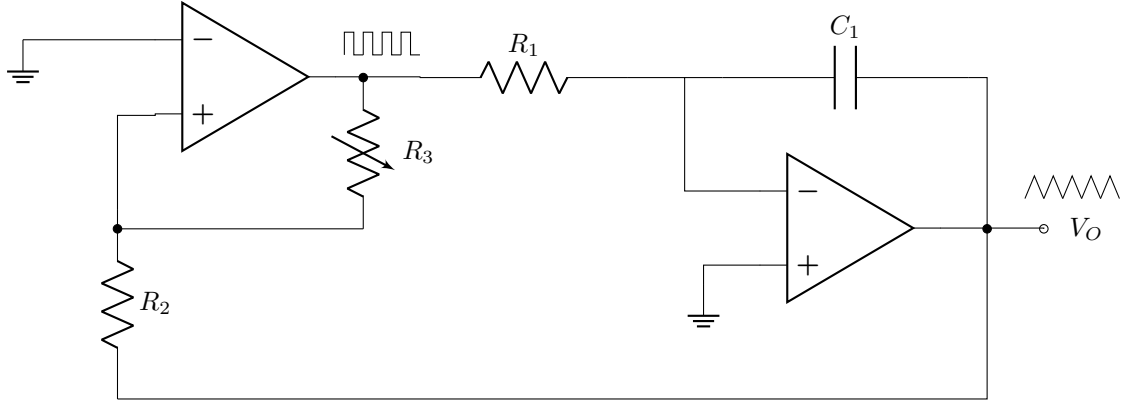


Figure 2: Circuit Diagram of Triangular Waveform Generator

2.1.1 Design for Required Specifications

Given: $f = 20$ kHz and $V_{o(pp)} = 12$ V (For Triangular Waves).

Assume the Op-Amp saturation voltage as $V_{sat} = \pm 7.5$ V.

For amplitude,

$$V_{o(pp)} = 2 \frac{R_2}{R_3} V_{sat}$$

Let's take R_3 as 10 k Ω and substituting values,

$$12 = 2 \frac{R_2}{10} \times 7.5$$

Therefore, $R_2 = 8$ k Ω .

Choose standard values as $R_2 = 8.2$ k Ω

For frequency,

$$f = \frac{R_3}{4R_1CR_2}$$

Substituting known values,

$$20 \times 10^3 = \frac{10 \times 10^3}{4 \cdot R_1 \cdot C \cdot 8.2 \times 10^3}$$

$$R_1C = 1.5244 \times 10^{-5}$$

Choose $C = 10$ nF.

Then,

$$R_1 = \frac{1.5244 \times 10^{-5}}{10 \times 10^{-9}} = 1524.4 \Omega$$

Choose the nearest standard value: $R_1 = 1.5$ k Ω .

Final component values are:

$R_1 = 1.5$ k Ω , $R_2 = 8.2$ k Ω , $R_3 = 10$ k Ω and $C = 10$ nF.

2.2 Buck Converter Design - Basic Calculations

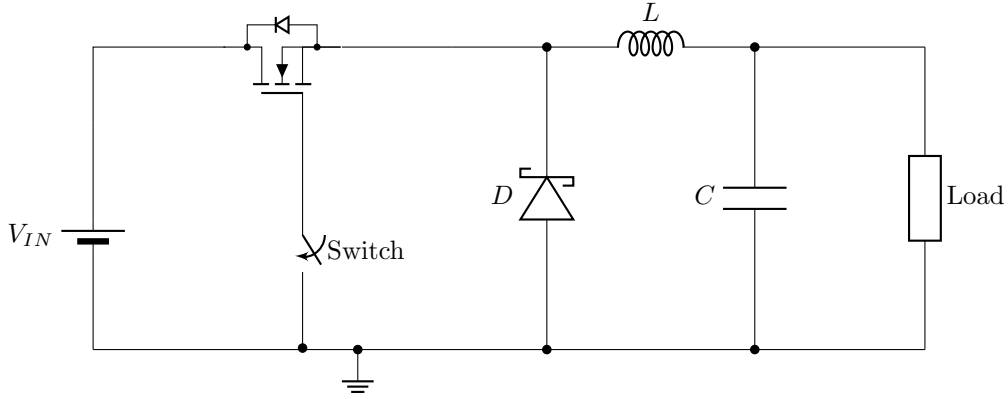


Figure 3: Circuit Diagram

Buck converter equations relate input voltage (V_{in}), output voltage (V_{out}), duty cycle (D), switching frequency (f_s) and component values (L , C) to achieve step-down DC-DC conversion. These equations are typically derived assuming Continuous Conduction Mode (CCM).

2.2.1 Calculations

Target Inductor Ripple Current (ΔI_L): As a rule of thumb, we can choose 10% of output current.

$$\Delta I_L = 10\% \times I_{out} = 0.1 \times 1 = 0.1 \text{ A}$$

Target Inductor Ripple Voltage Also, as a rule of thumb, we can take the Voltage Ripple (ΔV_{out}) as 1% of output voltage.

$$\Delta V_{out} = 1\% \times 12 = 0.12 \text{ V}$$

Duty Cycle Calculation:

$$D = \frac{V_{out}}{V_{in}}$$

$$D_{min} = \frac{3}{15} = 0.2, \quad D_{max} = \frac{12}{15} = 0.8$$

Inductor Calculation (worst case at $V_{out} = 12 \text{ V}$):

$$L = \frac{V_{out}(1 - D)}{f_s \cdot \Delta I_L}$$

$$L = \frac{12(1 - 0.8)}{18000 \times 0.1} = \frac{12 \times 0.2}{1800} = 0.00133 \text{ H} = 1.33 \text{ mH}$$

Capacitor Calculation:

$$C = \frac{\Delta I_L}{8f_s \cdot \Delta V_{out}}$$

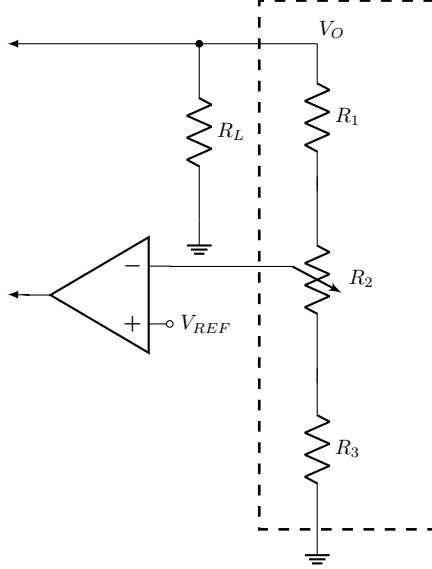
$$C = \frac{0.1}{8 \times 18000 \times 0.03} = 23.148 \text{ }\mu\text{F}$$

Peak Inductor Current:

$$I_{L,peak} = I_{out} + \frac{\Delta I_L}{2} = 1 + \frac{0.1}{2} = 1.05 \text{ A}$$

Note

All formulae above assume an ideal buck converter (no losses, ideal components and operation in Continuous Conduction Mode).

2.3 Feedback Voltage Divider

To achieve the target output voltage range of 3.0 V to 12.0 V using a 2.5 V reference voltage (V_{REF}), the feedback resistor network parameters were established. Based on a fixed potentiometer value of 100 k Ω for R_2 , the required values for the upper limiting resistor (R_1) and lower biasing resistor (R_3) were determined to ensure proper scaling across the full adjustment range.

$R_1 = 26.67 \text{ k}\Omega \approx 27 \text{ k}\Omega$ and $R_3 = 33.33 \text{ k}\Omega \approx 33 \text{ k}\Omega$ were selected.

Figure 4: Feedback Voltage Divider Circuit Diagram

2.4 2.5V Reference Voltage Generation

To provide a stable reference voltage of 2.5 V for the system, an LM431 adjustable shunt regulator is utilized. Because the target output voltage matches the internal reference voltage ($V_{ref} = 2.5 \text{ V}$), the device is configured without an external feedback voltage divider by directly shorting the reference (REF) pin to the cathode node.

Operating from a 15 V input supply rail, a 5.6 k Ω current-limiting bias resistor (R_{cl}) was selected. The nominal quiescent current (I_K) flowing through the regulator is determined by Ohm's Law as shown in Equation 1:

$$I_K = \frac{V_{in} - V_{out}}{R_{cl}} = \frac{15 \text{ V} - 2.5 \text{ V}}{5.6 \text{ k}\Omega} \approx 2.23 \text{ mA} \quad (1)$$

This design choice ensures that the bias current remains safely above the LM431's minimum required operating current ($I_{min} = 1 \text{ mA}$) to maintain tight regulation, while keeping overall power dissipation minimal.

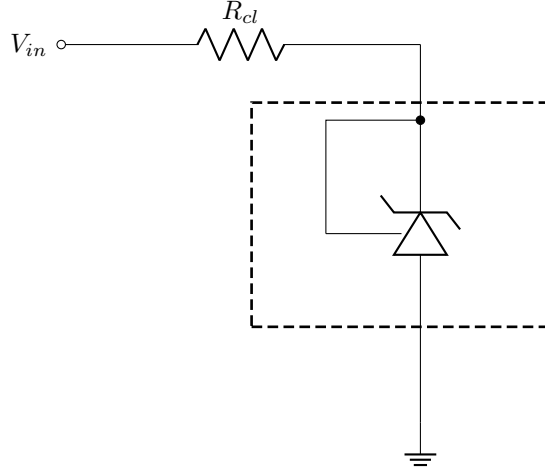


Figure 5: LM431 configured for a fixed 2.5V output from a 15V rail.

3 Detailed Design

3.1 Average Small Signal Switch Modelling

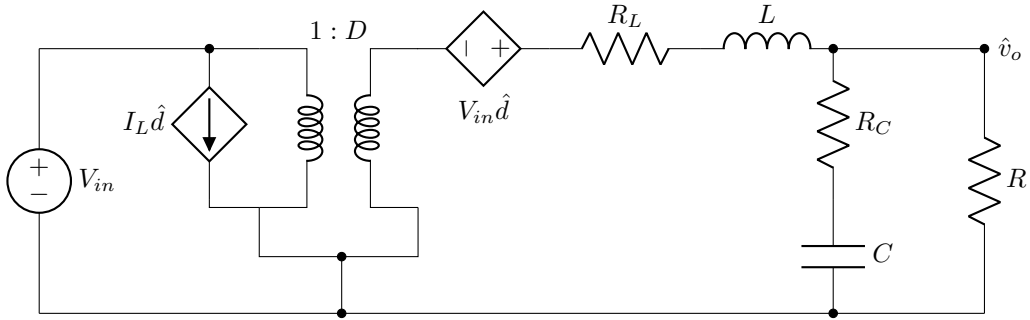


Figure 6: Small Signal Average Model of three Terminal PWM Switch

The control to output transfer function, $\frac{\hat{v}_o(s)}{\hat{d}(s)}$ is given by,

$$\frac{\hat{v}_o(s)}{\hat{d}(s)} = V_{in} \cdot \frac{1 + \frac{s}{\omega_z}}{1 + \frac{s}{Q\omega_0} + \left(\frac{s}{\omega_0}\right)^2} \quad (2)$$

$$Q \approx \frac{R}{\sqrt{\frac{L}{C}}}, \quad \omega_z = \frac{1}{R_C C}, \quad \omega_0 \approx \frac{1}{\sqrt{LC}}$$

We have chosen $C = 470 \mu\text{F}$, $L = 10 \text{ mH}$, and R_C measured as

$$f_{LC} = \frac{1}{2\pi\sqrt{10 \times 10^{-3} \times 470 \times 10^{-6}}} \approx \boxed{73.412 \text{ Hz}}$$

$$f_{z,ESR} = \frac{1}{2\pi \cdot R_{ESR} \cdot C} = \frac{1}{2\pi \times 0.2 \times 470 \times 10^{-6}} \approx \boxed{1693 \text{ Hz}}$$

The Power Stage transfer function $G_{\text{plant}}(s)$ can be written as,

$$G_{\text{plant}}(s) = F_m \cdot \frac{\hat{v}_o(s)}{\hat{d}(s)} \quad (3)$$

Where $F_m = \frac{1}{V_{pp}}$

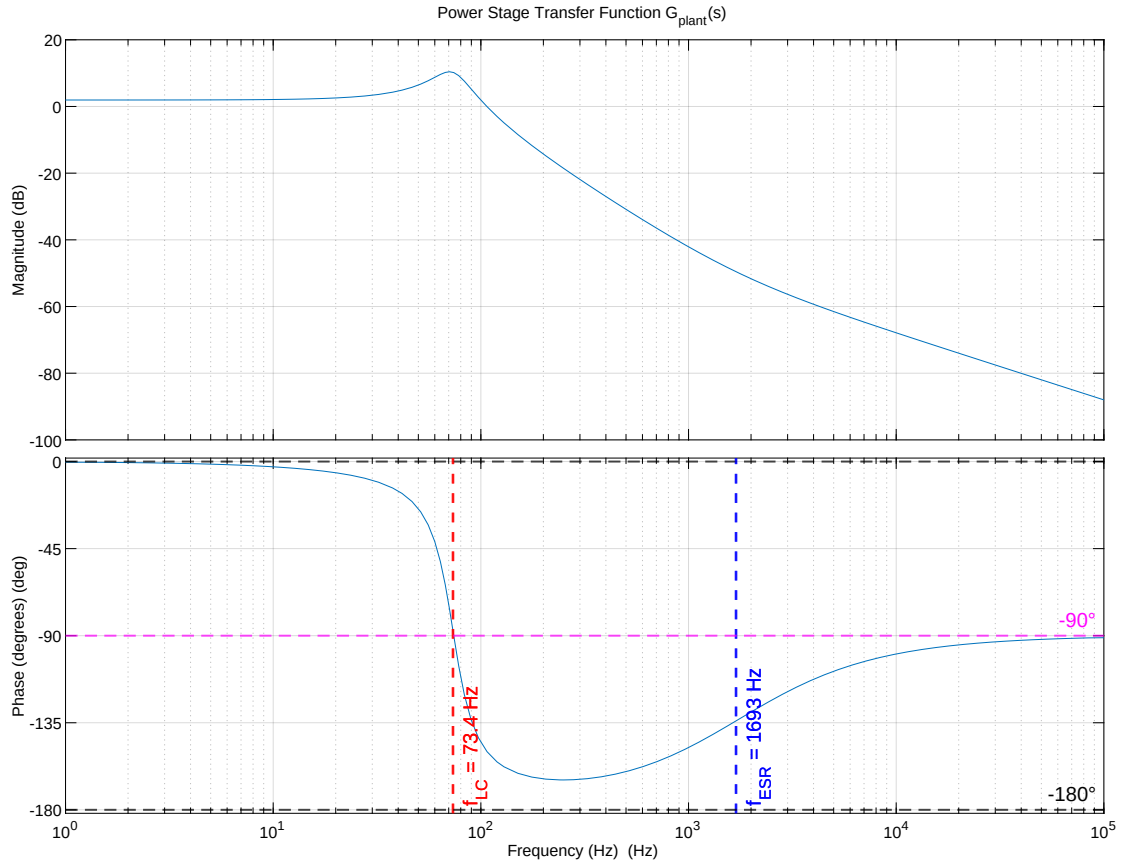


Figure 7: Power Stage Bode Diagram

3.2 Compensator Design

Compensator transfer function is given by,

$$G_c(s) = -\frac{1}{sR_1C_1} \cdot \frac{s + \frac{1}{R_2C_2}}{s + \frac{1}{R_2C_1}} \quad (4)$$

The compensator introduces:

- A zero at $\omega_z = -\frac{1}{R_2C_2}$
- A pole at $\omega_p = -\frac{1}{R_2C_1}$
- An integrator pole at DC (ensures zero steady-state error)

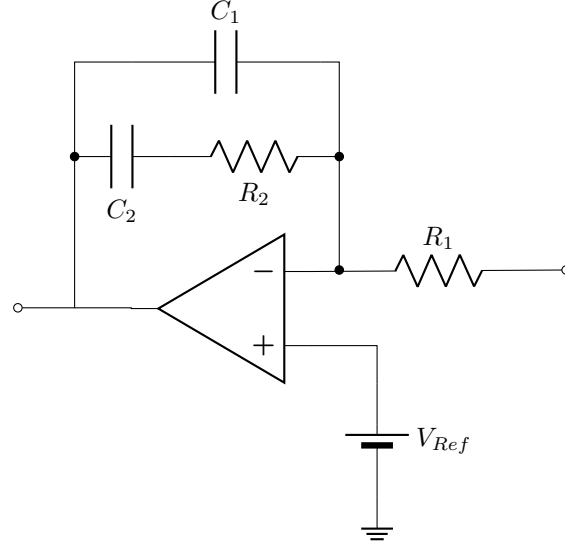


Figure 8: Type II compensator

Closed-loop requirements:

- cross-over frequency: $f_c = 1000$ Hz ($< 0.1f$) Phase Margin: $PM = 60^\circ$

$$\omega_c = 2\pi f_c = 6283.185 \text{ rad/s}$$

At the cross-over frequency,

$$PM - 180^\circ = \angle F(j\omega_c) + \angle G_c(j\omega_c) + \angle M(j\omega_c)$$

$$\angle M(j\omega_c) = 0 \text{ and } \angle F(j\omega_c) = -145^\circ \Rightarrow \angle G_c(j\omega_c) = 25^\circ$$

$$|F(j\omega_c)G_c(j\omega_c)M(j\omega_c)| = 1$$

. Therefore, $|F(j\omega_c)| = 98.42$

Using **K-Factor** Method, we can find,

$$K = \tan\left(\frac{\angle G_c(j\omega_c)}{2}\right) = 0.2217$$

Then

$$\omega_z = \omega_c \left(\sqrt{\frac{1-K}{1+K}} \right) \quad \omega_p = \omega_c \left(\sqrt{\frac{1+K}{1-K}} \right)$$

Then, $\omega_z = 5015$ rad/s and $\omega_p = 7872.06$ rad/s. Taking $R_2 = 22 \text{ k}\Omega$, $C_1 = 10 \text{ nF}$ and $C_2 = 5.6 \text{ nF}$. R_1 is a part of feedback voltage divider described in section 2.3.

4 Component Selection

The selection of components is based on the design requirements of a switching frequency (f_s) of 18 kHz, an output current of 1 A and an input voltage range up to 15 V. The following components were selected based on their performance, availability and compatibility with high-speed switching.

4.1 Power MOSFET: IRF3205

The IRF3205 was selected as the main switching element. The primary criteria for selection were low ON-resistance ($R_{DS(on)}$) and high current handling.

- **Voltage Rating:** It has a Drain-to-Source breakdown voltage (V_{DSS}) of 55 V, which provides a safety margin of over 200% for our 15 V input.
- **Current Rating:** Capable of handling up to 110 A continuous drain current. Our peak current is only 1.05 A, ensuring the device operates far below its thermal limits.
- **Efficiency:** With an $R_{DS(on)}$ of only 8 m Ω , the conduction power loss is negligible:

$$P_{cond} = I_{out}^2 \times R_{DS(on)} = (1)^2 \times 0.008 = 8 \text{ mW}$$

4.2 MOSFET Driver: IR2110

Driving a High-Side N-channel MOSFET requires a gate voltage higher than the input voltage ($V_{in} + V_{GS}$). The IR2110 is a high-voltage, high-speed power MOSFET driver with independent high and low-side referenced output channels.

- **Floating Channel:** It uses a bootstrap circuit to provide the necessary gate-to-source voltage for the high-side IRF3205.
- **Switching Speed:** It offers typical rise/fall times of 25ns/17ns. This is critical for minimizing switching losses at 20 kHz, ensuring the MOSFET transitions quickly through the linear region.
- **Drive Current:** With a peak output current of 2 A, it can rapidly charge the gate capacitance (C_{iss}) of the IRF3205.

4.3 Operational Amplifier: NE5532

For the triangular wave generator and the error amplifier (compensator), the NE5532 was chosen over standard op-amps like the LM741 or TL082.

- **Slew Rate:** It features a high slew rate of 9 V/ μ s. This is essential for generating a clean 20 kHz triangular wave without “rounding” the peaks.
- **Bandwidth:** A unity-gain bandwidth of 10 MHz ensures that the control loop responds quickly to load transients.
- **Low Noise:** As a high-performance audio op-amp, it provides low input noise, which is beneficial for the stability of the PWM comparator.

4.4 Voltage Reference: LM431

To maintain a stable output voltage, the feedback loop requires a precise reference voltage (V_{REF}).

- **Precision:** The LM431 provides a programmable output reference of 2.5 V with high thermal stability.
- **Regulation:** It replaces a simple Zener diode because it offers a much sharper “turn-on” knee and lower dynamic impedance, ensuring the control loop compares the output against a rock-steady baseline despite input voltage fluctuations.

4.5 Freewheeling Diode: 1N5819 Schottky Barrier Rectifier

The **1N5819** Schottky diode was selected to provide the freewheeling path during the MOSFET off-state based on the following justifications:

- **Negligible Reverse Recovery ($t_{rr} \approx 0$ ns):** Unlike slow silicon P-N diodes (e.g., 1N4007), the metal-to-silicon Schottky junction switches instantly at 18 kHz, eliminating reverse recovery power losses and voltage ringing.
- **Low Forward Voltage Drop ($V_F \approx 0.45$ V):** The low drop reduces worst-case diode conduction losses ($D_{min} = 0.2$) to just 0.36 W, a 59% reduction in power dissipation compared to a standard silicon diode (0.88 W).
- **Safe Operating Margins:** The diode's 40 V reverse breakdown rating provides a $2.67\times$ safety factor against the 15 V input voltage rail and its 1.0 A average forward current rating perfectly matches the maximum 1 A load requirement.

5 Software Simulation

5.1 Triangle Waveform Generation

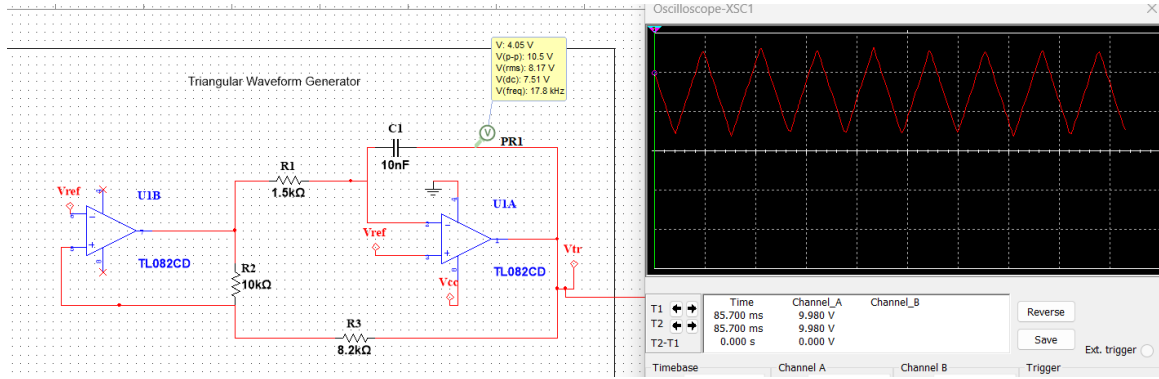


Figure 9: Simulation Results of Triangle waveform Generator (Using NI Multisim)

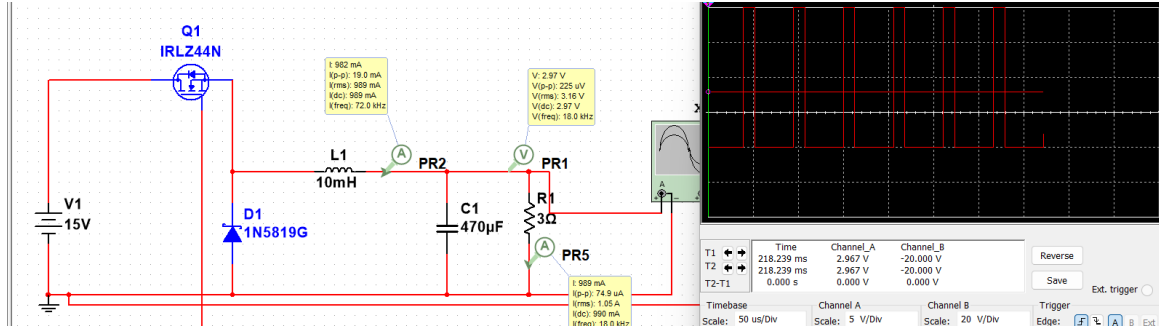
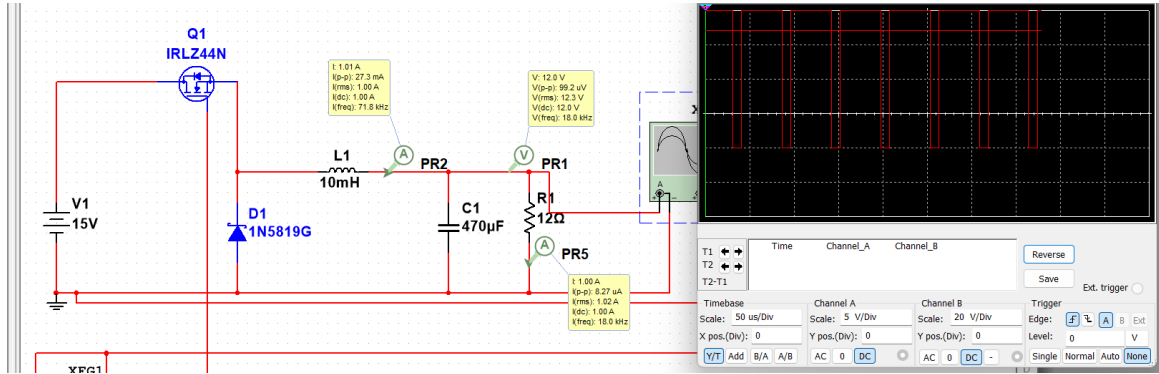
Table 1: Comparison of Theoretical and Simulated Values for Triangular Waveform Generator

Parameter	Theoretical Value	Simulated Value	Percentage Error (%)
Frequency (f)	20.0 kHz	17.8 kHz	11.0%
Peak-to-Peak (V_{pp})	12.0 V	10.5 V	12.5%
Center Voltage (V_{dc})	7.5 V	7.52 V	0.27%

5.2 Buck Converter Simulations

To validate the theoretical design of the buck converter, the ideal calculated values were compared against simulation results obtained at a switching frequency of 18 kHz. Table presents the side-by-side comparison of the inductor current ripple (ΔI_L) and output voltage ripple (ΔV_{out}) for both the 3 V and 12 V operating conditions.

Output	Parameter	Theoretical	Simulated
3 V	Current Ripple (ΔI_L)	13.33 mA	19.0 mA
	Voltage Ripple (ΔV_{out})	197 μ V	225 μ V
12 V	Current Ripple (ΔI_L)	13.33 mA	27.3 mA
	Voltage Ripple (ΔV_{out})	197 μ V	99.2 μ V

Figure 10: Simulation Results for $V_O=3$ VFigure 11: Simulation Results for $V_O=12$ V

The mathematical expressions for an ideal buck converter dictate identical ripple magnitudes at both operating points because the product $V_{out}(1 - D)$ evaluates to a constant 2.4 V. However, the simulated data reveals noticeable variances:

- **Inductor Current Ripple:** The simulation yields higher current ripples (19.0mA and 27.3mA) than the ideal baseline. This shift is primarily attributed to non-ideal switches, diode forward voltage drops, and internal inductor DC resistance (DCR), which alter the effective voltage across the inductor (V_L) during the conduction phases.
- **Output Voltage Ripple:** At 3 V, the simulated voltage ripple increased to 225 μ V, a typical consequence of introducing Capacitor Equivalent Series Resistance (ESR). Conversely, the lower voltage ripple observed at 12 V (99.2 μ V) indicates dampening effects within the simulation loop, potentially influenced by higher continuous damping losses or software solver convergence constraints at high duty cycles.

6 Analysis & Discussion

6.1 Triangle Waveform Generation

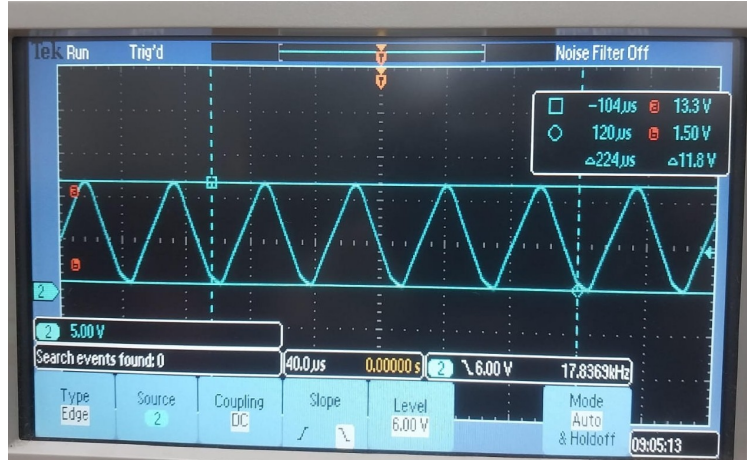


Figure 12: Triangular Waveform Output

The total power dissipation (P) of the triangular waveform generator is calculated using the supply voltage (V) and the load current (I) as shown in Equation 5.

$$P = V \times I = 15 \text{ V} \times 6 \text{ mA} = 90 \text{ mW} \quad (5)$$

Table 2: Updated Performance and Operating Parameters of the Generator

Parameter	Practical Value	Theoretical Value	Percentage Error (%)
Frequency (f)	17.8 kHz	20 kHz	11.00%
Peak-to-Peak (V_{pp})	11.8 V	12 V	1.67%
Center Voltage (V_{dc})	7.4 V	7.5 V	1.33%

6.2 Reference Voltage Stability

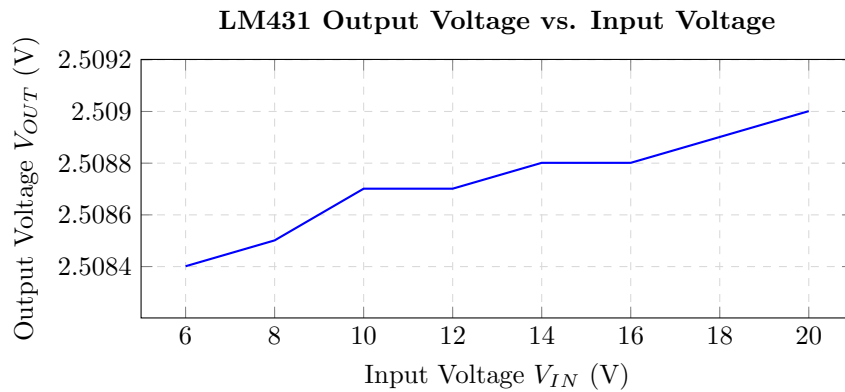


Figure 13: LM431 Line Regulation Performance Plot

6.3 Line Regulation Analysis

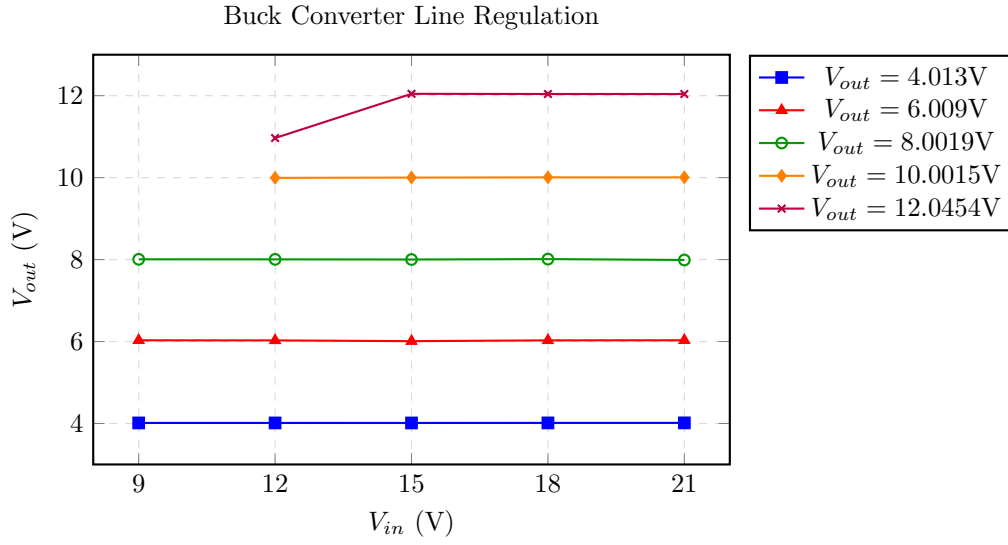


Figure 14: Line Regulation: Output Voltage vs. Input Voltage.

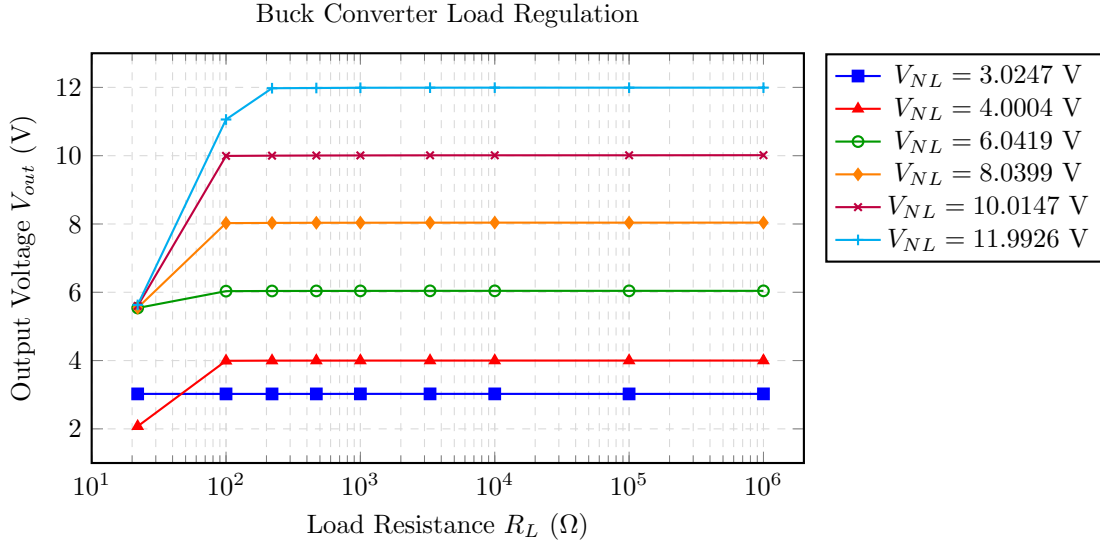
Table 3: Line Regulation Data Table (In Volts)

	V_{in}				
V_{out}	9	12	15	18	21
4.013	4.0135	4.0133	4.013	4.0143	4.0149
6.009	6.0305	6.028	6.009	6.0292	6.031
8.0019	8.0064	8.0057	8.0019	8.012	7.9908
10.0015		9.9944	10.0015	10.0067	10.0063
12.0454		10.966	12.0454	12.0408	12.04

6.4 Load Regulation Analysis

Table 4: Load Regulation Performance Under Various Loads

Load	V_{out} Without Load (V)					
Resistance	3.0247	4.0004	6.0419	8.0399	10.0147	11.9926
22 Ω	3.0239	2.0755	5.54	5.547	5.6	5.6302
100 Ω	3.0236	3.9933	6.0332	8.0254	9.9934	11.06
220 Ω	3.0231	3.9983	6.0377	8.0313	10.0008	11.9762
470 Ω	3.0237	3.9994	6.0394	8.0354	10.0054	11.9823
1 kΩ	3.0246	3.9997	6.0405	8.0365	10.0079	11.9892
3.3 kΩ	3.0246	4.0001	6.0418	8.0382	10.011	11.9917
10 kΩ	3.0243	4.0002	6.042	8.0394	10.0115	11.9928
100 kΩ	3.0234	4.0003	6.0418	8.0389	10.012	11.9921
1 MΩ	3.0247	4.0004	6.0419	8.0399	10.0147	11.9926

Figure 15: Load Regulation Graph: Output Voltage vs. Load Resistance (R_L).

6.5 Load Current Analysis

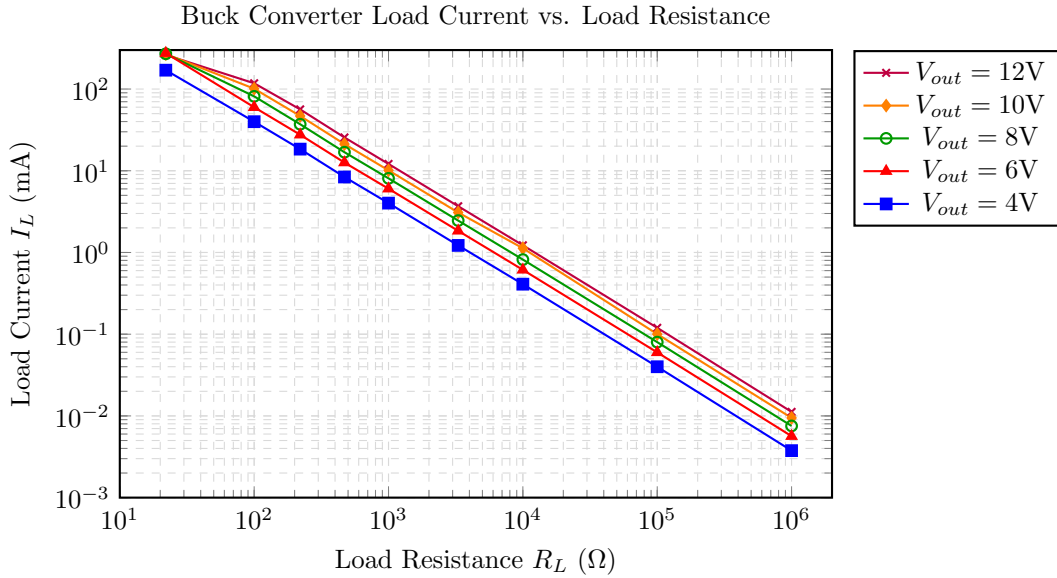


Figure 16: Load Current (mA) vs. Load Resistance across different target output voltages.

6.6 Converter Efficiency Analysis

Note on Quiescent Auxiliary Overhead: A permanent base supply operating current of 9 mA (135 mW power) flows through the circuit under absolute no-load conditions. This constant residual overhead is consumed continuously by active signal conditioning elements, including the PWM triangle carrier wave ramp generator, high-speed comparative latch blocks, and error amplifier compensation loops. As a result, when operating at light loads ($R_L \geq 10 \text{ k}\Omega$), this internal circuit overhead dominates total energy consumption, leading to a sharp drop toward zero efficiency.

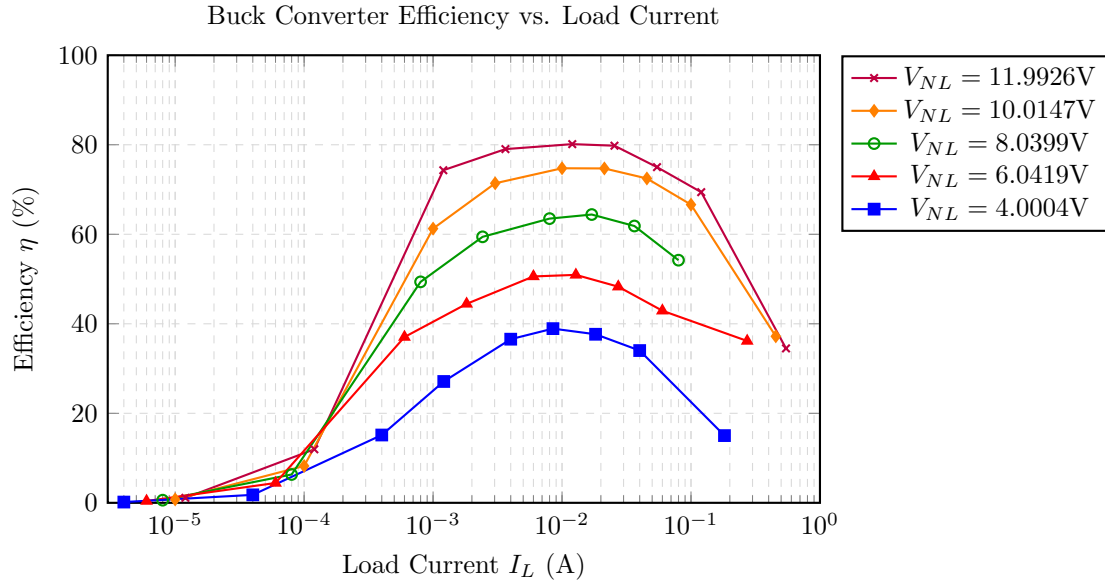


Figure 17: Converter Efficiency (%) versus Load Current (I_L) at $V_{in} = 15$ V.

6.7 Power Stage Efficiency Analysis

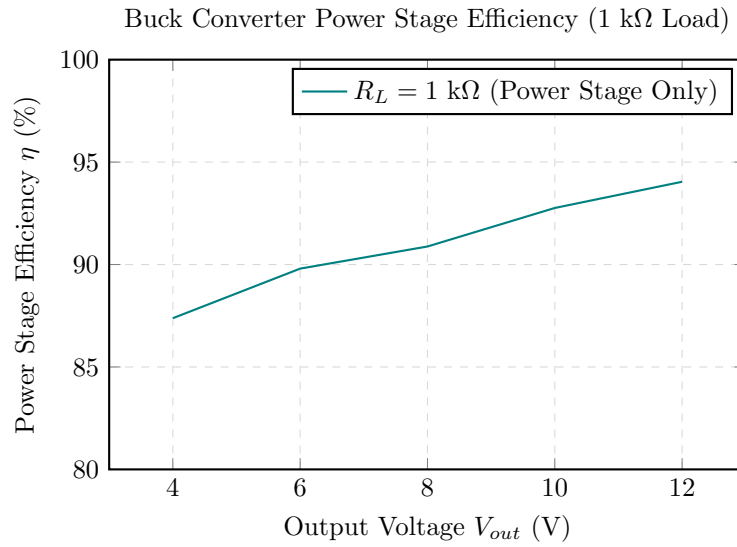


Figure 18: Power stage conversion efficiency vs. output voltage ($V_{in} = 15$ V).

6.8 Diode Voltage Waveforms in a Buck Converter



Figure 19: Diode Voltage Waveforms in a Buck Converter

6.9 Output Voltage Waveform

The output voltage (V_{out}) of the buck converter was measured across the load capacitor using a digital storage oscilloscope. The resulting experimental waveform is displayed in Figure 20.

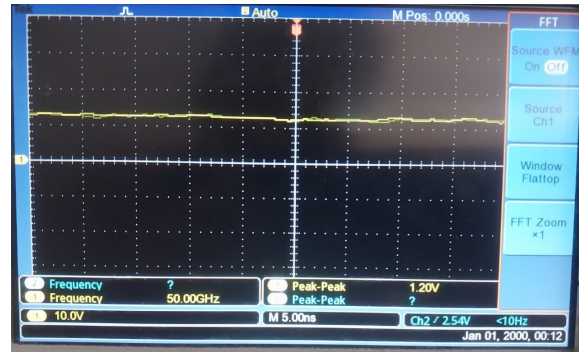


Figure 20: Oscilloscope capture of the buck converter output voltage (V_{out}).

The frequency measurement reads an anomalous 50.00 GHz. This occurs because the ripple signal is so small and smooth that the oscilloscope's hardware counter is triggering on background random noise rather than the actual switching frequency.

7 Challenges and Future Improvements

7.1 Main Technical Challenges

1. Limitations in Switching Frequency Realization

While standard switch-mode power supplies (SMPS) typically operate in the 100 kHz to 500 kHz range to optimize passive component sizing, this fully analog implementation was constrained to a maximum stable switching frequency of 20 kHz. Extensive experimental iterations were conducted using high-slew-rate operational amplifiers, including the TL072, TL082 etc; however, due to real-world bandwidth degradation, non-ideal component tolerances, and severe parasitic characteristics found in locally sourced IC batches, these devices failed to stably generate the required high-frequency carrier wave.

Consequently, the NE5532 dual low-noise operational amplifier was utilized as the structural workaround. The resulting low switching frequency introduced substantial downstream hard-

were constraints, primarily making it exceptionally difficult to source high-inductance components capable of handling the rated power stage saturation currents within the local market. This bottleneck ultimately forced the deployment of a physically over-sized, bulky inductor core, compromising the power density of the converter.

2. High Quiescent Analog Power Dissipation

Because the entire control topology—including the triangle carrier wave oscillator, error amplifier feedback networks, and PWM comparator blocks—was implemented using discrete, active analog sub-circuits, a continuous and substantial power overhead was observed. Under absolute no-load conditions ($I_L = 0$ A), the control circuitry maintains a static baseline current draw of 9 mA directly from the 15 V main supply. This fixed 135 mW power consumption severely hurts the overall system efficiency profile, particularly when evaluating operation under light-load or high-resistance conditions, where the auxiliary control losses completely dominate the input power envelope.

7.2 Proposed Future Improvements

To mitigate the structural drawbacks identified during empirical evaluation, the following design improvements are proposed for future revisions of the system:

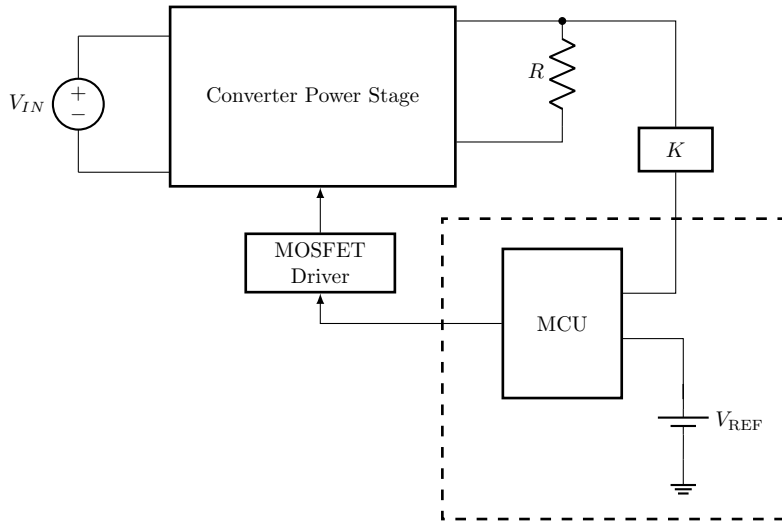


Figure 21: Mixed-Signal Architecture Block Diagram

To replace the continuous-time analog error amplifier with the digital architecture shown in Fig. 21, the controller is implemented as a discrete-time recursive difference equation.

References

- [1] J. G. Kassakian, M. F. Schlecht, and G. C. Verghese, *Principles of Power Electronics*. Reading, MA: Addison-Wesley, 1991. ISBN: 9780201096897.
- [2] Texas Instruments, *Buck Converter Modeling, Control and Compensator Design*
- [3] EEEguide, “Triangular Wave Generator using Op-amp,” [Online]. Available: <https://www.eeeguide.com/triangular-wave-generator-using-op-amp/>.
- [4] EPCI Passive Components Europe, “Buck Converter Design and Calculation,” *Passive Components Magazine*, [Online]. Available: <https://passive-components.eu/buck-converter-design-and-calculation/>.

A Overall Circuit Diagram

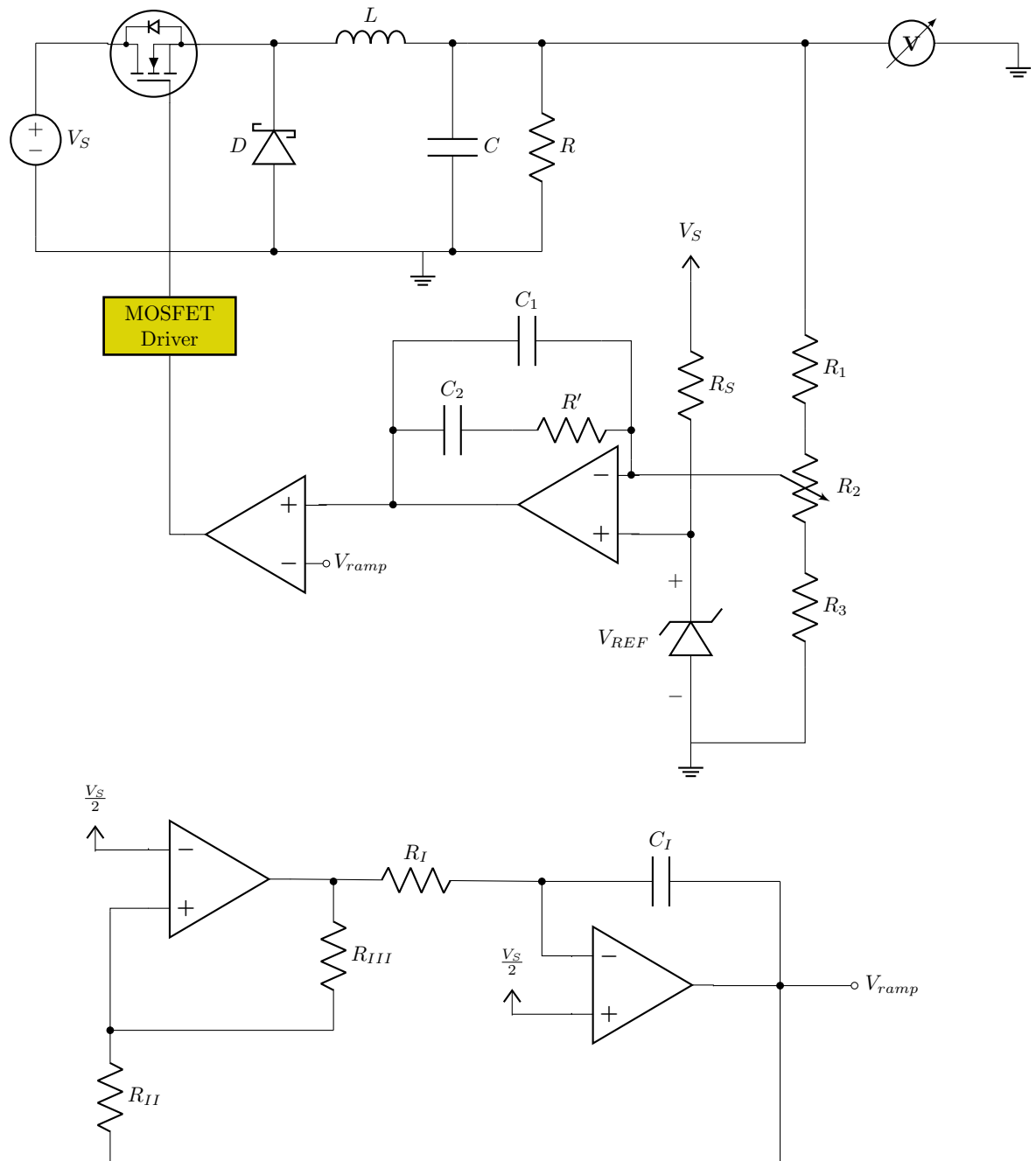


Figure 22: Overall Circuit Diagram